

Accounting Business and Society

–Performance Measurement

WEEK10

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Key concepts

- 01** Decentralisation & Responsibility Centres
- 02** Performance measurement: ROI vs Residual Income
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1# Decentralization in Organizations: Benefits

Benefits of Decentralization

Lower-level decisions often based on better information.

Lower-level managers gain experience in decision-making.

Top management freed to concentrate on strategy.

Lower level managers can respond quickly to customers.

Decision-making authority leads to job satisfaction.

1# Decentralization in Organizations: Disadvantages

Lower-level managers may make decisions without seeing the “big picture.”

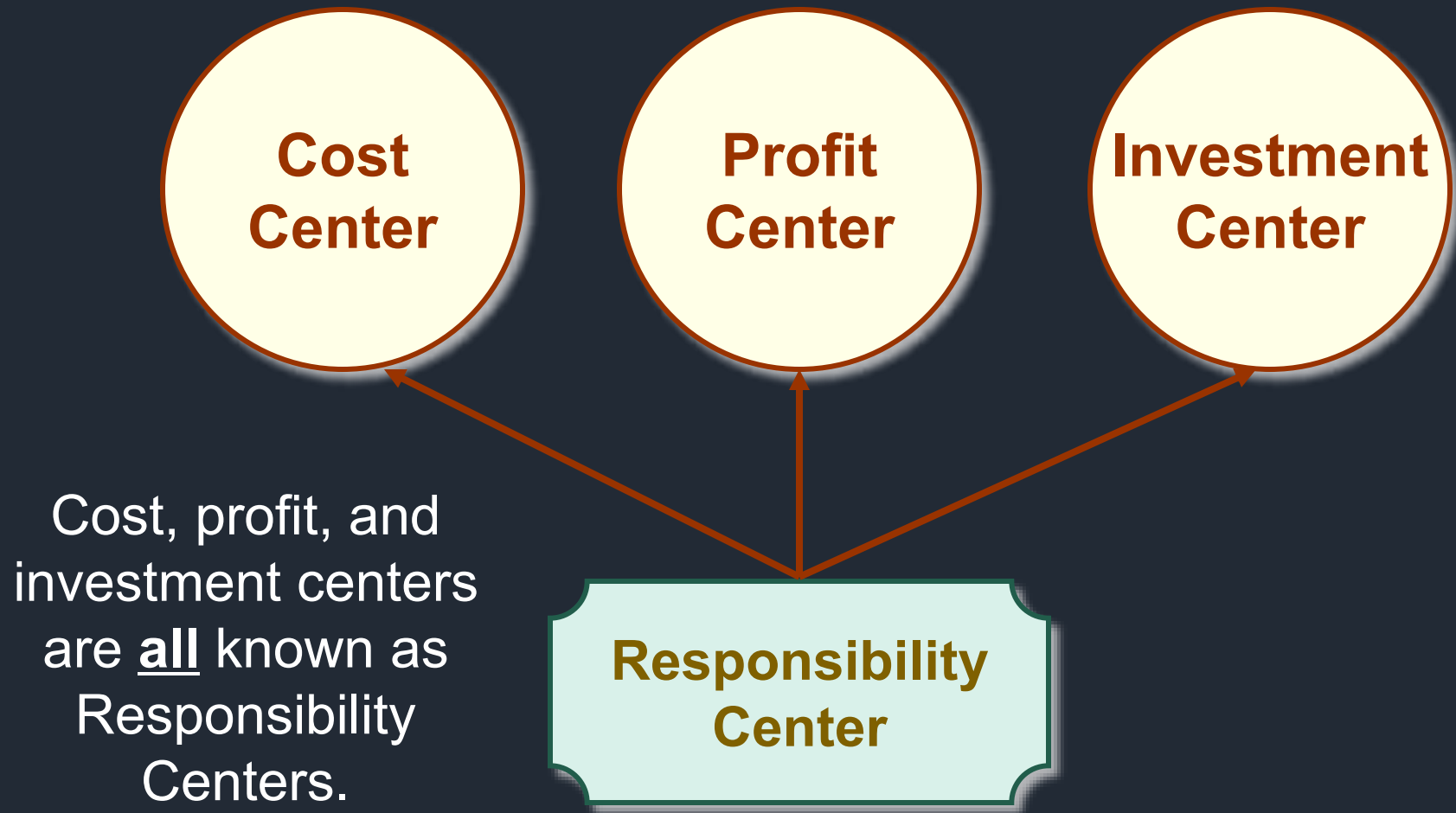
May be a lack of coordination among autonomous managers.

Lower-level manager's objectives may not be those of the organization.

Disadvantages of Decentralization

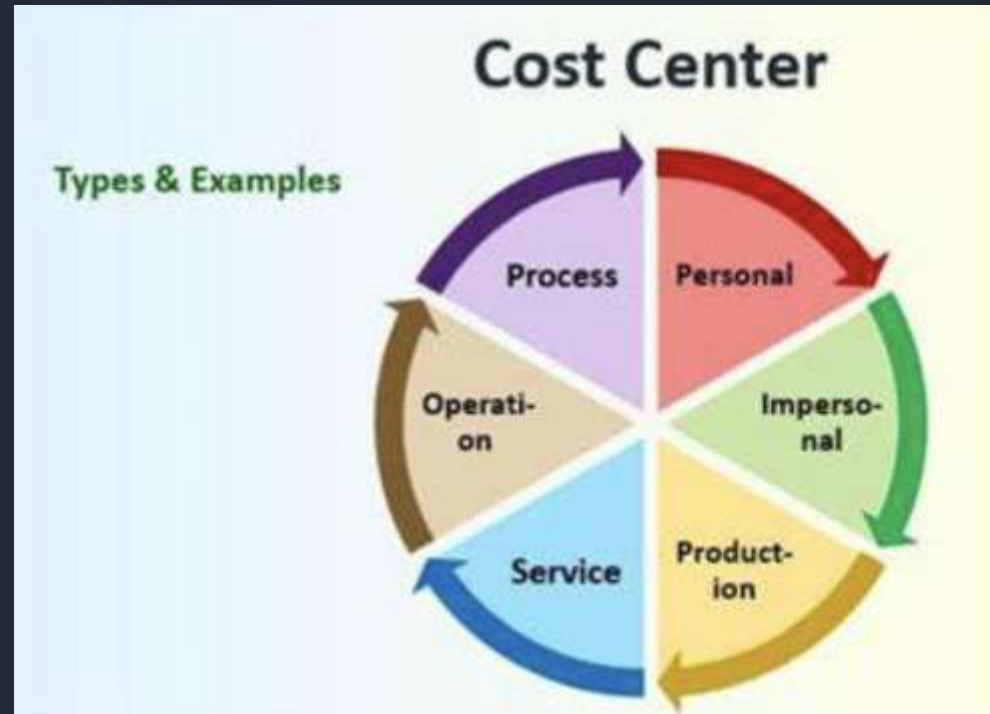
May be difficult to spread innovative ideas in the organization.

Responsibility Accounting



Cost Center

A segment whose manager has control over costs, but not over revenues or investment funds.



Profit Center

A segment whose manager has control over **both** costs and revenues, but no control over investment funds.

Revenues

Sales

Interest

Other

Costs

Mfg. costs

Commissions

Salaries

Other

Investment Center

A segment whose manager has control over costs, revenues, and investments in operating assets.

**investment
center**



2# Performance measurement: **ROI** vs Residual Income

Income before interest
and taxes (EBIT)

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Average operating assets}}$$

Cash, accounts receivable, inventory,
plant and equipment, and other
productive assets.

Net Book Value versus Gross Cost

Most companies use the net book value of depreciable assets to calculate average operating assets.

Acquisition cost
Less: Accumulated depreciation
Net book value

Understanding ROI

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Average operating assets}}$$

$$\text{Margin} = \frac{\text{Net operating income}}{\text{Sales}}$$

$$\text{Turnover} = \frac{\text{Sales}}{\text{Average operating assets}}$$

$$\text{ROI} = \text{Margin} \times \text{Turnover}$$

Increasing ROI: An Example

Regal Company reports the following:

Net operating income	\$ 30,000
Average operating assets	\$ 200,000
Sales	\$ 500,000
Operating expenses	\$ 470,000

What is Regal Company's ROI?

$$\text{ROI} = \text{Margin} \times \text{Turnover}$$

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Average operating assets}}$$

Increasing ROI - An Example: Solution

$$\text{ROI} = \text{Margin} \times \text{Turnover}$$

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Average operating assets}}$$

$$\text{ROI} = \frac{\$30,000}{\$500,000} \times \frac{\$500,000}{\$200,000}$$

$$\text{ROI} = 6\% \times 2.5 = 15\%$$

Investing in Operating Assets to Increase Sales - An Example

Assume that Regal's manager invests in a \$30,000 piece of equipment that increases sales by \$35,000, while increasing operating expenses by \$15,000.

Regal Company reports the following:

Net operating income	\$ 50,000
Average operating assets	\$ 230,000
Sales	\$ 535,000
Operating expenses	\$ 485,000

Let's calculate the new ROI.

Investing in Operating Assets to Increase Sales - An Example: Solution

$$\text{ROI} = \text{Margin} \times \text{Turnover}$$

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Average operating assets}}$$

$$\text{ROI} = \frac{\$50,000}{\$535,000} \times \frac{\$535,000}{\$230,000}$$

$$\text{ROI} = 9.35\% \times 2.33 = 21.8\%$$

ROI increased from 15% to 21.8%.

Criticisms of ROI

In the absence of the balanced scorecard, management may not know how to increase ROI.

Managers often inherit many committed costs over which they have no control.

Managers evaluated on ROI may reject profitable investment opportunities.

2# Performance measurement: ROI vs Residual Income

Residual Income is net operating income above some minimum return on operating assets.



Calculating Residual Income

$$\text{Residual income} = \text{Net operating income} - \left(\text{Average operating assets} \times \text{Minimum required rate of return} \right)$$

This computation differs from ROI.

ROI measures net operating income earned relative to the investment in average operating assets.

Residual income measures net operating income earned less the minimum required return on average operating assets.

Residual Income – An Example

The Retail Division of Zephyr, Inc. has average operating assets of \$100,000 and is required to earn a return of 20% on these assets.

In the current period, the division earns \$30,000.

Let's calculate residual income.

Residual Income – An Example: Solution

Operating assets	\$ 100,000
Required rate of return ×	20%
Minimum required return	<u><u>\$ 20,000</u></u>

Actual income	\$ 30,000
Minimum required return	<u>(20,000)</u>
Residual income	<u><u>\$ 10,000</u></u>

Motivation and Residual Income

Residual income encourages managers to make profitable investments that would be rejected by managers using ROI.

Quick Check 1

Redmond Awnings, a division of Wrap-up Corp., has a net operating income of \$60,000 and average operating assets of \$300,000. The required rate of return for the company is 15%. What is the division's ROI?

- a. 25%
- b. 5%
- c. 15%
- d. 20%

Quick Check 1a

Redmond Awnings, a division of Wrap-up Corp., has a net operating income of \$60,000 and average operating assets of \$300,000. The required rate of return for the company is 15%. What is the division's ROI?

- a. 25%
- b. 5%
- c. 15%
- d. 20%

$$\begin{aligned}\text{ROI} &= \text{NOI} / \text{Average operating assets} \\ &= \$60,000 / \$300,000 = 20\%\end{aligned}$$

Quick Check 2

Redmond Awnings, a division of Wrap-up Corp., has a net operating income of \$60,000 and average operating assets of \$300,000. If the manager of the division is evaluated based on ROI, will she want to make an investment of \$100,000 that would generate additional net operating income of \$18,000 per year?

- a. Yes
- b. No

Quick Check 2a

Redmond Awnings, a division of Wrap-up Corp., has a net operating income of \$60,000 and average operating assets of \$300,000. If the manager of the division is evaluated based on ROI, will she want to make an investment of \$100,000 that would generate additional net operating income of \$18,000 per year?

a. Yes

☒ b. No

$$\text{ROI} = \$78,000 / \$400,000 = 19.5\%$$

This lowers the division's ROI from 20.0% down to 19.5%.

Quick Check 3

The company's required rate of return is 15%. Would the company want the manager of the Redmond Awnings division to make an investment of \$100,000 that would generate additional net operating income of \$18,000 per year?

- a. Yes
- b. No

Quick Check 3a

The company's required rate of return is 15%. Would the company want the manager of the Redmond Awnings division to make an investment of \$100,000 that would generate additional net operating income of \$18,000 per year?

- a. Yes
- b. No

$$\text{ROI} = \$18,000 / \$100,000 = 18\%$$

The return on the investment exceeds the minimum required rate of return.

Quick Check 4

Redmond Awnings, a division of Wrap-up Corp., has a net operating income of \$60,000 and average operating assets of \$300,000. The required rate of return for the company is 15%. What is the division's residual income?

- a. \$240,000
- b. \$ 45,000
- c. \$ 15,000
- d. \$ 51,000

Quick Check 4a

Redmond Awnings, a division of Wrap-up Corp., has a net operating income of \$60,000 and average operating assets of \$300,000. The required rate of return for the company is 15%. What is the division's residual income?

- a. \$240,000
- b. \$ 45,000
- ☒ c. \$ 15,000
- d. \$ 51,000

Net operating income	\$60,000
Required return (15% of \$300,000)	<u>(45,000)</u>
Residual income	<u>\$15,000</u>

Quick Check 5

If the manager of the Redmond Awnings division is evaluated based on residual income, will she want to make an investment of \$100,000 that would generate additional net operating income of \$18,000 per year?

- a. Yes
- b. No

Quick Check 5a

If the manager of the Redmond Awnings division is evaluated based on residual income, will she want to make an investment of \$100,000 that would generate additional net operating income of \$18,000 per year?

- ☒ a. Yes
- ☐ b. No

Net operating income	\$78,000
Required return (15% of \$400,000)	<u>(60,000)</u>
Residual income	<u>\$18,000</u>

Yields an increase of \$3,000 in the residual income.

Divisional Comparisons and Residual Income

The residual income approach has one major disadvantage.

It cannot be used to compare the performance of divisions of different sizes.

Zephyr, Inc. – Part 1

Recall the following information for the Retail Division of Zephyr, Inc.

Assume the following information for the Wholesale Division of Zephyr, Inc.

Operating assets

Required rate of return ×

Minimum required return

Retail

\$ 100,000

20%

\$ 20,000

Wholesale

\$ 1,000,000

20%

\$ 200,000

Actual income

Minimum required return

Residual income

Retail

\$ 30,000

(20,000)

\$ 10,000

Wholesale

\$ 220,000

(200,000)

\$ 20,000

Zephyr, Inc. – Part 2

The residual income numbers suggest that the Wholesale Division outperformed the Retail Division because its residual income is \$10,000 higher. However, the Retail Division earned an ROI of 30% compared to an ROI of 22% for the Wholesale Division. The Wholesale Division's residual income is larger than the Retail Division **simply because it is a bigger division**.

	Retail	Wholesale
Operating assets	\$ 100,000	\$ 1,000,000
Required rate of return ×	20%	20%
Minimum required return	\$ 20,000	\$ 200,000
	Retail	Wholesale
Actual income	\$ 30,000	\$ 220,000
Minimum required return	(20,000)	(200,000)
Residual income	\$ 10,000	\$ 20,000

3# Segmented Income Statement

Understand how to
construct and use a
balanced scorecard.

Decentralization and Segment Reporting

A **segment** is any part or activity of an organization about which a manager seeks cost, revenue, or profit data.

An Individual Store

A Service Center

A Sales Territory

Keys to Segmented Income Statements

There are two keys to building segmented income statements:

A contribution format should be used because it separates fixed from variable costs and it enables the calculation of a contribution margin.

Traceable fixed costs should be separated from common fixed costs to enable the calculation of a segment margin.

Identifying Traceable Fixed Costs

Traceable fixed costs arise because of the existence of a particular segment and would disappear over time if the segment itself disappeared.

Examples of traceable fixed costs include:

The salary of the Fritos product manager at PepsiCo is a traceable fixed cost of the Fritos business segment of PepsiCo.

The maintenance cost for the building in which Boeing 747s are assembled is a traceable fixed cost of the 747 business segment of Boeing.

Identifying Common Fixed Costs

Common fixed costs arise because of the overall operation of the company and would not disappear if any particular segment were eliminated.

Examples of common fixed costs include:

The salary of the CEO of General Motors is a common fixed cost of the various divisions of General Motors.

The cost of heating a Safeway or Kroger grocery store is a common fixed cost of the various departments.

Traceable Fixed Costs

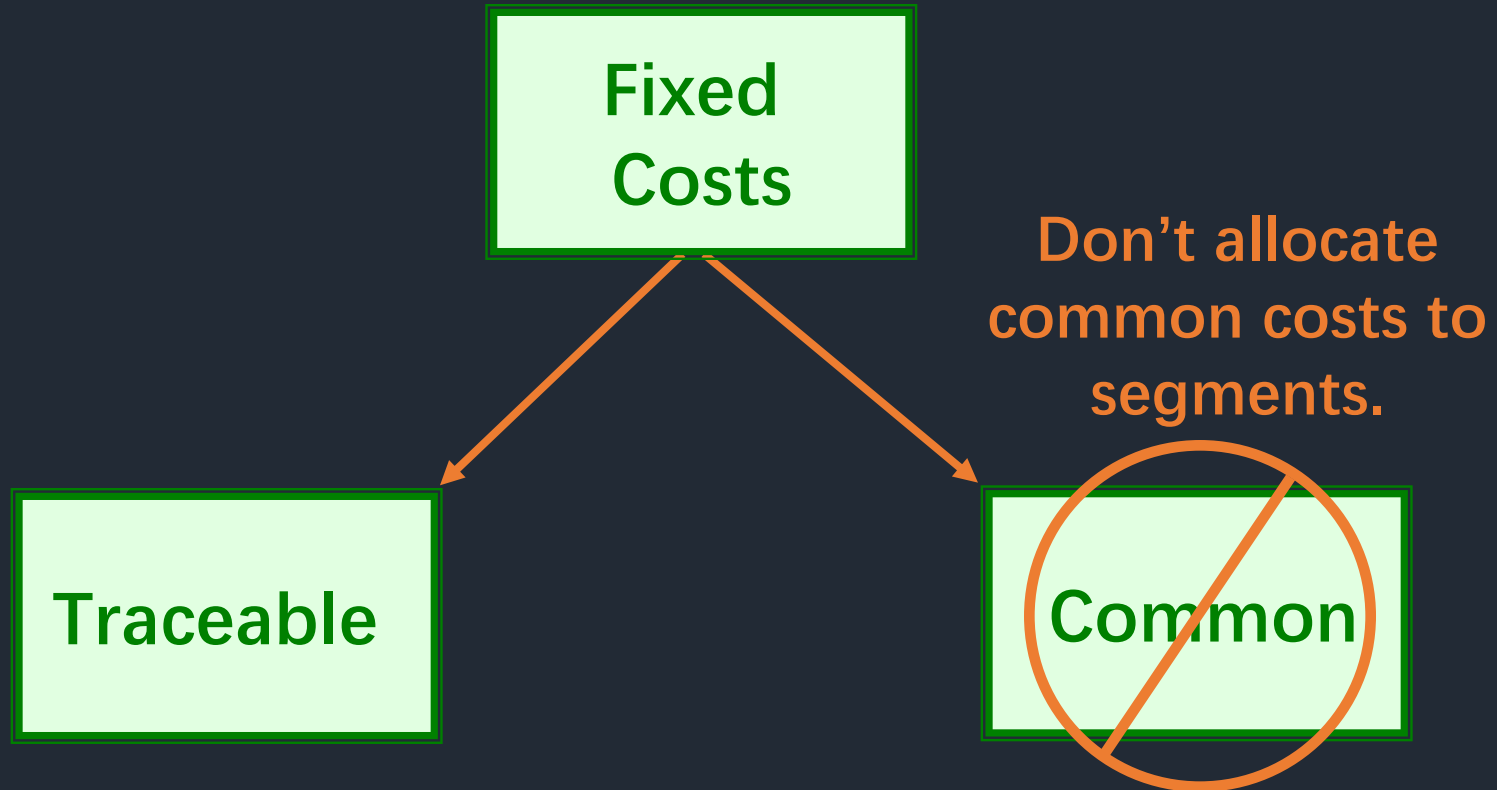
It is important to realize that the traceable fixed costs of one segment may be a common fixed cost of another segment.

For example, the landing fee paid to land an airplane at an airport is traceable to the particular flight, but it is not traceable to first-class, business-class, and economy-class passengers.

Segment Margin

The **segment margin**, which is computed by subtracting the traceable fixed costs of a segment from its contribution margin, is the **best gauge** of the long-run profitability of a segment.

Traceable and Common Costs



Levels of Segmented Statements – Part 1

Webber, Inc. has two divisions.

- 1. Computer Division**
- 2. Television Division**

**Let's look more closely at the
Television Division's income
statement.**

Levels of Segmented Statements - Part 4

Income Statement

	Company	Television	Computer
Sales	\$ 500,000	\$ 300,000	\$ 200,000
Variable expenses	230,000	150,000	80,000
CM	270,000	150,000	120,000
Traceable FC	170,000	90,000	80,000
Division margin	100,000	\$ 60,000	\$ 40,000
Common expenses			
Net operating income			

Levels of Segmented Statements - Part 5

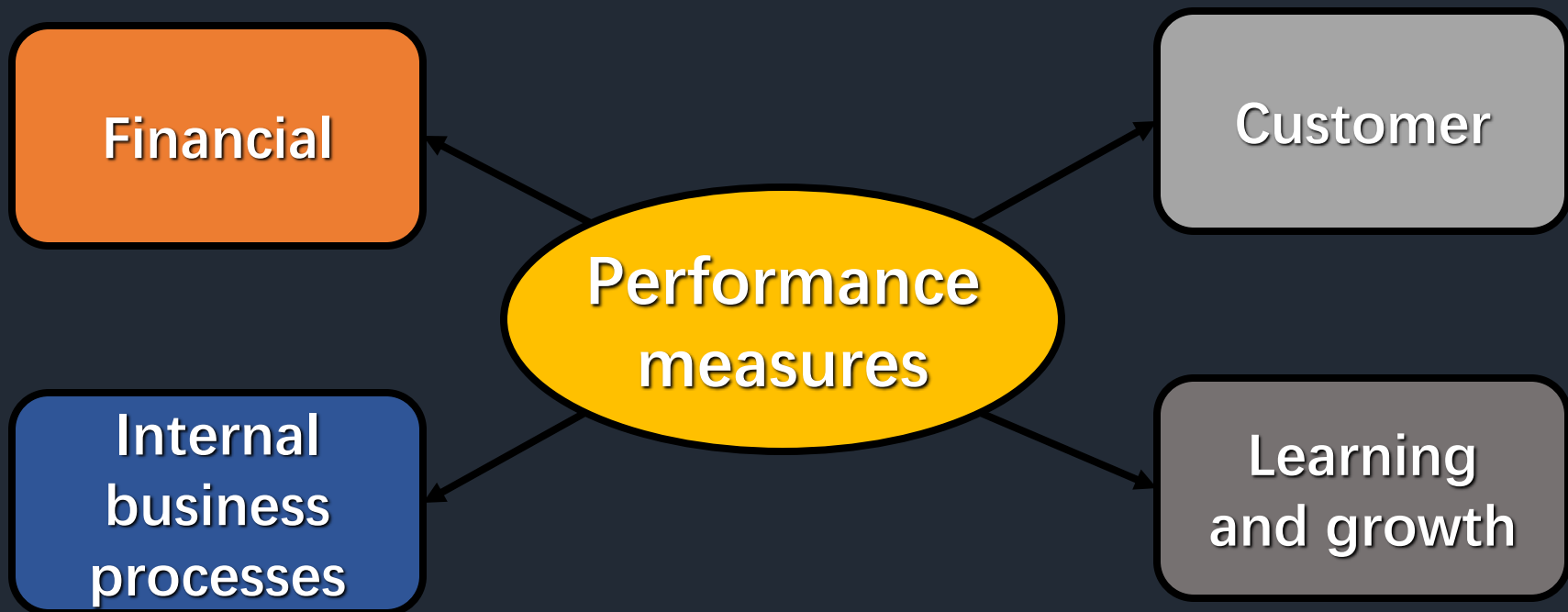
Income Statement

	Company	Television	Computer
Sales	\$ 500,000	\$ 300,000	\$ 200,000
Variable expenses	230,000	150,000	80,000
CM	270,000	150,000	120,000
Traceable FC	170,000	90,000	80,000
Division margin	100,000	\$ 60,000	\$ 40,000
Common expenses	25,000		
Net operating income	\$ 75,000		

Common fixed expenses should **not** be allocated to the divisions. These expenses would remain even if one of the divisions were eliminated.

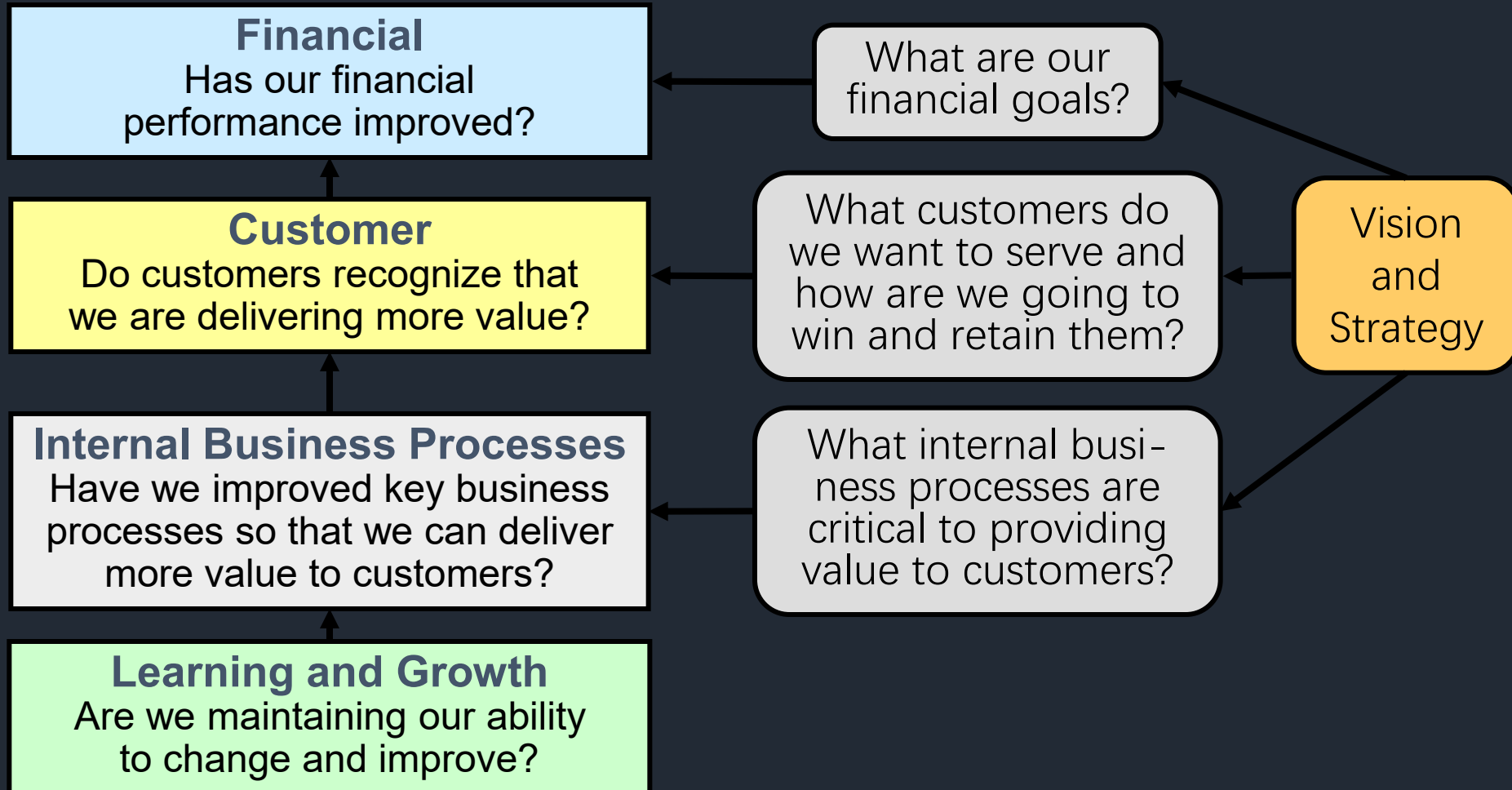
4# The Balanced Scorecard

Management translates its strategy into performance measures that employees understand and influence.



The Balanced Scorecard – From Strategy to Performance Measures

Performance Measures



The Balanced Scorecard – Non-financial Measures

The balanced scorecard relies on non-financial measures in addition to financial measures for two reasons:

➊ Financial measures are lag indicators that summarize the results of past actions. Non-financial measures are leading indicators of future financial performance.

➋ Top managers are ordinarily responsible for financial performance measures – not lower level managers. Non-financial measures are more likely to be understood and controlled by lower level managers.

The Balanced Scorecard for Individuals

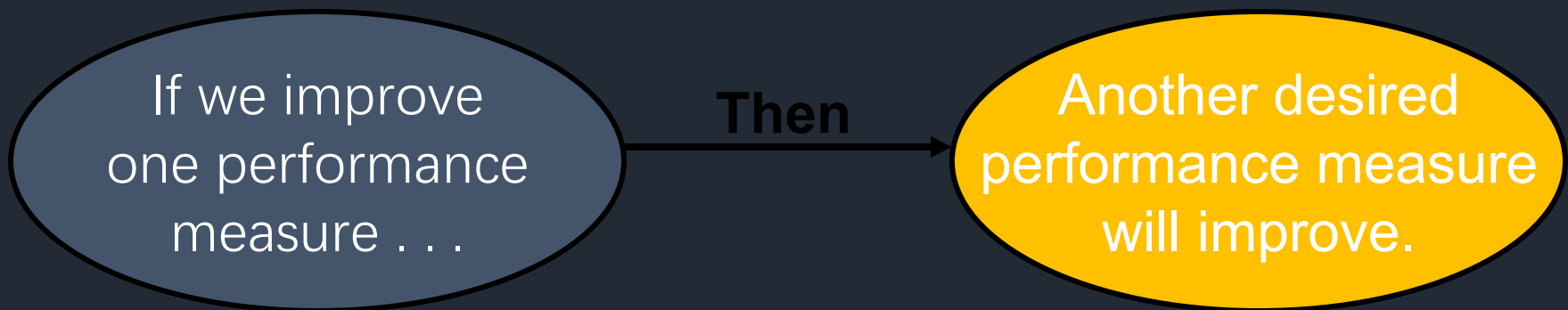
The entire organization should have an overall balanced scorecard.

Each individual should have a personal balanced scorecard.

A personal scorecard should contain measures that can be influenced by the individual being evaluated and that support the measures in the overall balanced scorecard.

The Balanced Scorecard – Important Links

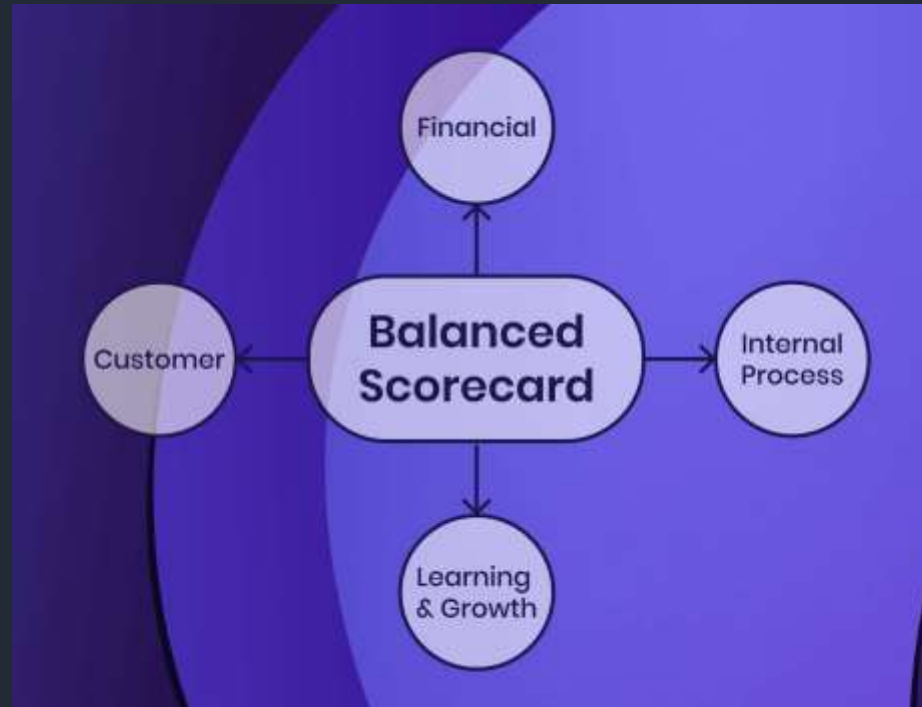
A balanced scorecard should have measures that are linked together on a cause-and-effect basis.



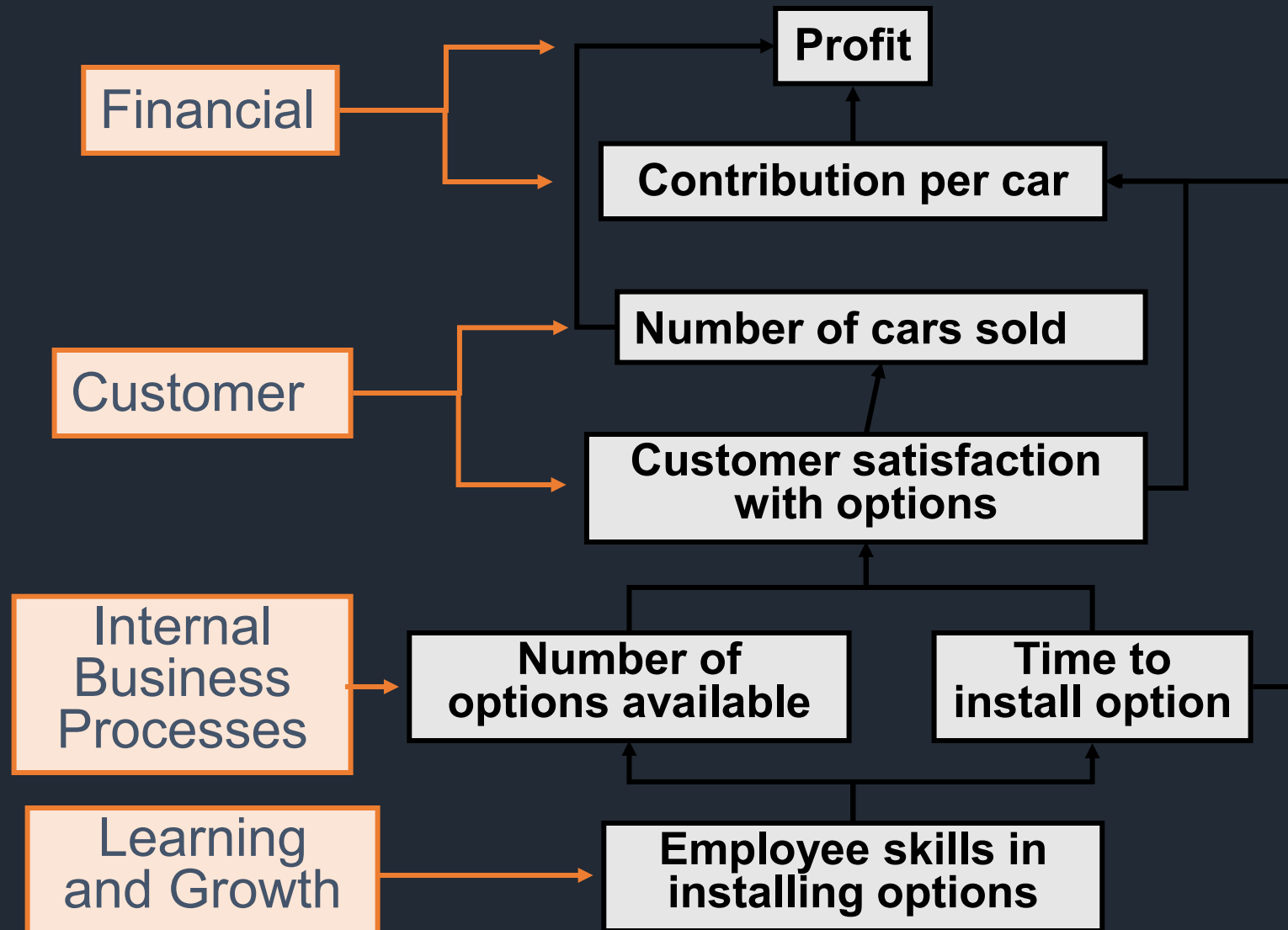
The balanced scorecard lays out concrete actions to attain desired outcomes.

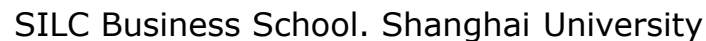
The Balanced Scorecard and Compensation

Incentive compensation should be linked to balanced scorecard performance measures.

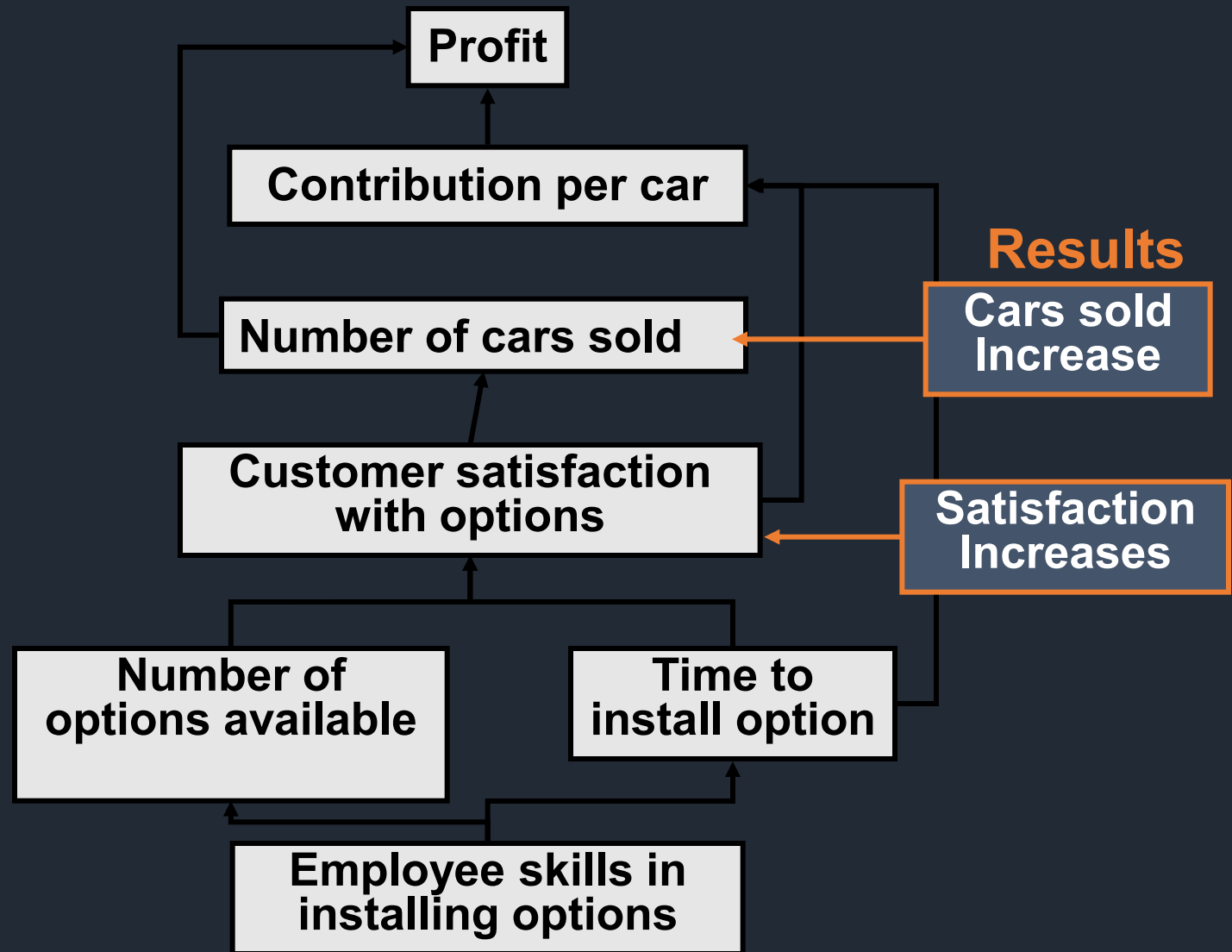


The Balanced Scorecard - Jaguar Example - Part 1

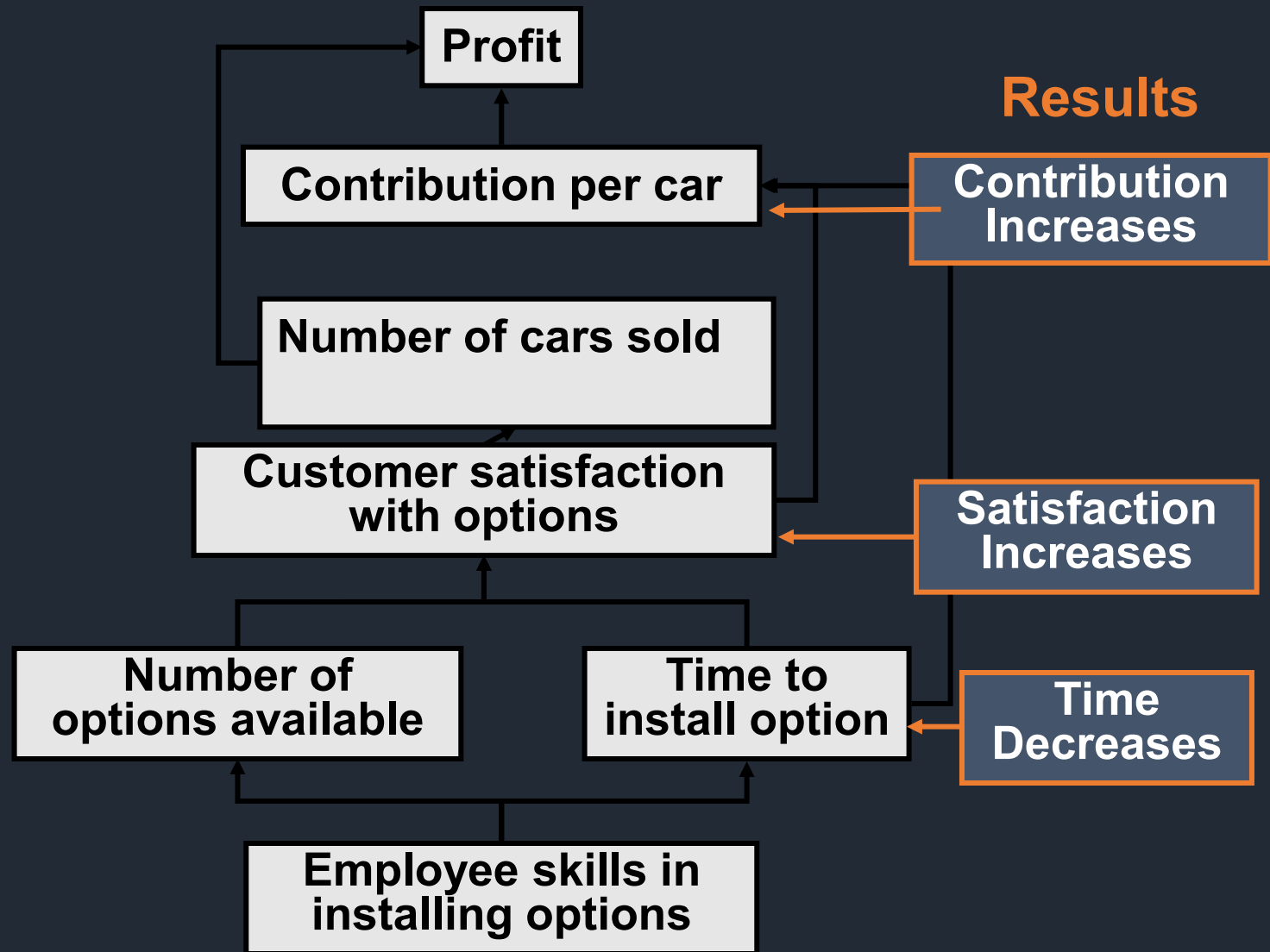




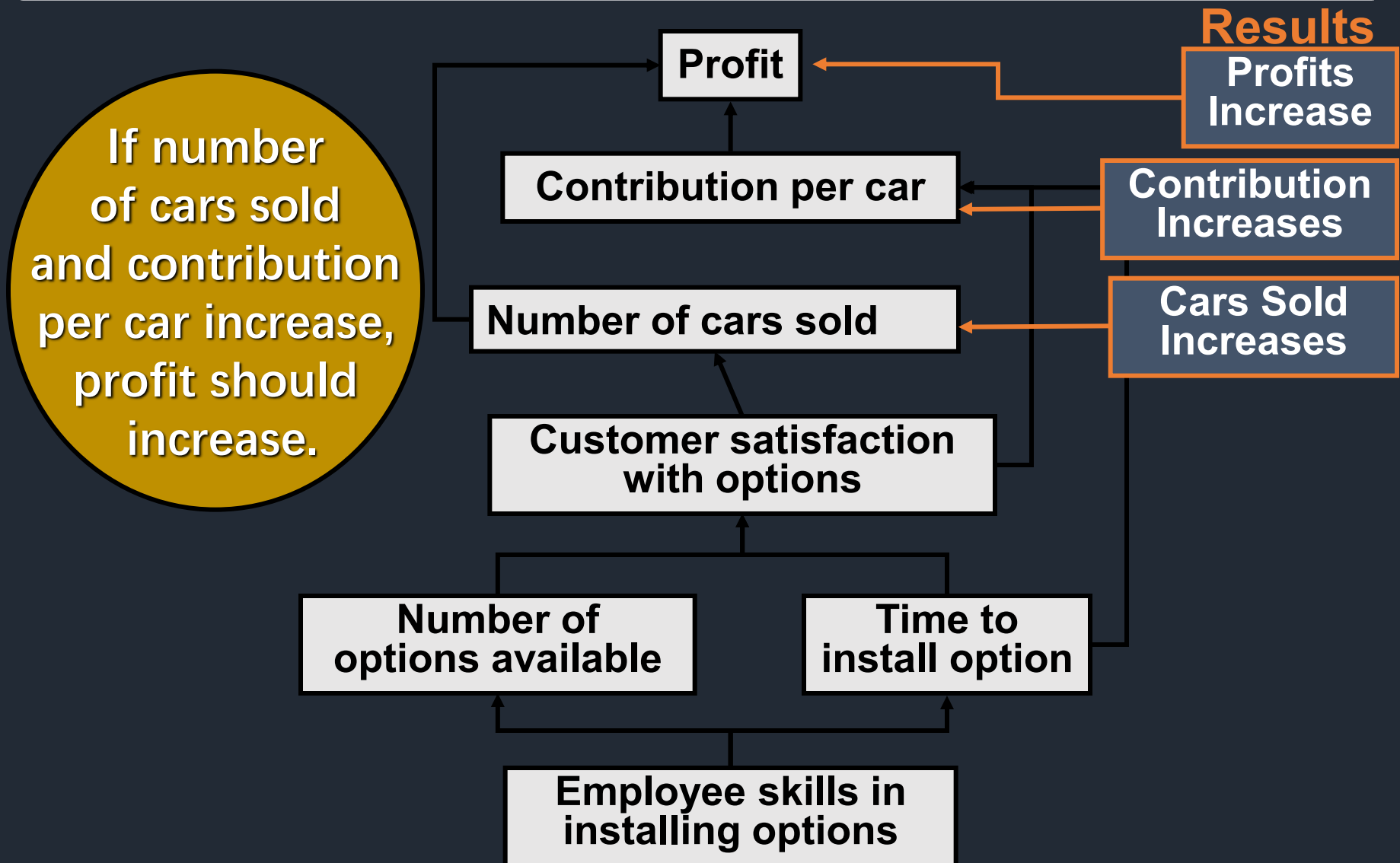
The Balanced Scorecard - Jaguar Example - Part 3



The Balanced Scorecard - Jaguar Example - Part 4



The Balanced Scorecard - Jaguar Example - Part 5



Assessment

Tutorial portfolio - Online quizzes:

This quiz is based on the week 10 learning materials. You may complete this anytime during week 10, you may use your study notes.

You have TWO attempts to complete this quiz.

Due time for your weekly quiz is on Canvas, please check it by yourself.